



Ensuring your water is suitable

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Know

Useful things to know

- Suitable water must be:
 - safe to drink if it is used for food and drink preparation, washing food contact surfaces/equipment, and for [staff](#) to wash their hands,
 - safe and fit for purpose when used for any other activities in growing or making food.

Why is it important to ensure water is suitable?

- Water can carry harmful [bugs](#) and chemicals that can make people sick. These might be because the water is [contaminated](#) at the source, or because water pipes and storage containers become contaminated.
- It is important to consider how you/your staff use water in your business, and make sure that the water is not going to contaminate your food or drink. If you use a registered water supply most of this is done for you.
- A registered drinking water supplier is someone who owns or operates a water supply and is responsible for making sure that it is safe. Suppliers have until November 2025 to register with Taumata Arowai. You can search for registered suppliers here: hinekorako.taumataarowai.govt.nz/publicregister/supplies/





- Dairy processors must follow the requirements for Water in C1.13 to C1.24 in the Animal Products Notice: Production, Supply and Processing www.mpi.govt.nz/dmsdocument/50182.



- Dairy: Wherever you get your water from (town supply or self-supply), you will need to:
 - develop water-use criteria,
 - do initial water testing and
 - perform routine monitoring.



- Dairy: You can do this by completing the Dairy Processors Template Water-use Plan (www.mpi.govt.nz/dmsdocument/59743), using the minimum requirements in the **Do** section. When this template is completed, it, combined with the Suitable Water Page, will be your water use plan and will include the water-use criteria. The Dairy Processors Template Water-use Plan does not cover all possible sources of water. If your water source is not covered, you will have to write your own water-use plan and water-use criteria. You could use the template water-use plan and the Suitable Water Page to help you do this.



If you use self-supply water

- You/your staff will need to prove it is safe for use by having it tested at an accredited lab. You can find a lab at: hinekorako.taumataarowai.govt.nz/publicregister/laboratories/.



- You/your staff will need to know what near-by activities and naturally occurring chemicals could make your water supply unsafe (for example nitrates for groundwater, or lead for roof water).

- Keep water tanks:
 - clean and in good condition to stop the build-up of sediment, and
 - covered to stop animals, birds and dirt from contaminating water.
-  Horticulture: What happens on the farm (or immediately post-harvest) can affect the safety and suitability of horticultural food. Unsafe water can contaminate food.
- You may need to install operate and maintain (for example replacing filters) a water treatment system, following the manufactures instructions, to ensure water is suitable for use with food and drink.
- You may need to treat roof, surface or ground water using filtration, chlorination or UV disinfection to make it suitable for use.
- Self-supply water sources may be subject to other legislation as well.

For ground water source only

- Bores should be designed and maintained so they are protected from surface contamination.

For roof water source only

- Additional risks to contamination of your water can be reduced by:
 - collecting water only from clean roofs and gutters made from safe materials (for example no lead based paints, bitumen, exposed timber or copper gutters),

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- putting screening gutters up, removing overhanging branches and vegetation, and mounting aerials and satellite dishes away from water collection areas,
- installing a first flush device (a device which diverts the first flush of water when it rains).



- Mobile businesses: You need to meet the following requirements for the source(s) of water for each of the different locations you operate in.



- Mobile businesses: You should **record** the water source for each of the locations you operate in.



- Horticulture: Water used for irrigation should be of drinkable quality where possible- especially where applied directly to crops that are not expected to be cooked or treated to reduce or eliminate bugs or chemicals, before being eaten.



- Wine: You will need to prove it is suitable for use by having it tested at an accredited and recognised lab.

D

Do

Rules you must follow

- For water for food processing, hand washing and cleaning, either:
 - use a registered drinking water supply (for example local council), or,
 - use a roof, surface or ground water supply that meets the requirements (that is ensure that it is safe).
- If using self-supply water you must test it if you do not have any testing **records** for it. It must meet the following limits:



Measurement	Criteria
Escherichia coli	Less than 1cfu/g in any 100 ml sample*
Turbidity	Must not exceed 5 Nephelometric Turbidity Units
Chlorine (when chlorinated)	Not less than 0.2mg/l (ppm) free available chlorine with a minimum of 30 minute contact time
pH (when chlorinated)	6.5 to 8.0

*Escherichia coli testing must be performed by an accredited lab.

- Test any new supply of water before using it in food areas.
- Wine: Self-supplied water must be tested at least once every year and within 1 week of any adverse event that may affect water quality.



- Wine: Test results must show self-supplied water meets the following limits:



Measurement	Criteria
Escherichia coli	Less than 1cfu/g in any 100 ml sample*
Turbidity	Must not exceed 5 Nephelometric Turbidity Units

*Escherichia coli testing must be performed by a laboratory accredited to ISO/IEC 17025 Testing and Calibration Laboratories with required tests listed or specified within the laboratory's scope of accreditation.

- For surface water sources, and ground water sources, water intakes must be:
 - at least 10m away from livestock,
 - at least 50m away from potential sources of contamination including silage stacks, offal pits, human and animal waste, potential chemical stores and tanks.

D

Do

- Identify any hazards, including naturally occurring (for example heavy metals and nitrates) that could be in your water supply and control these appropriately.

All water supplies

- Only use water tanks, containers, pipes, outlet taps and treatment systems for any water supplies on site that are safe for drinking water. Regularly check and maintain these.
- Clearly mark any water that is not suitable for use in food production including water for handwashing and surface cleaning.
- If your water supply becomes unsafe (or you are advised by your supplier it is unsafe), follow any instructions from your water supplier before using it. If they do not provide you with advice to follow:
 - do not use it, or
 - boil it for at least 1 minute before use, or
 - disinfect it with chlorine before use, or
 - use another supply of water which you are sure is safe (for example bottled water or water from a registered water supplier).
- Always throw out any food or drink which has been contaminated.



It is recommended that you **record** the water source for each of the locations you operate in.



- Dairy: Water is used for (tick all applicable):
 - ☐ cleaning of facilities and equipment
 - ☐ personal hygiene activities
 - ☐ production of steam or ice
 - ☐ ingredient or direct contact with product

☐ other operational activities where water comes into direct or indirect contact with any product

☐ other (specify) _____



Table 1: Standard requirements for water and routine testing frequency

Measurement	Maximum limit	Minimum routine testing frequency
Escherichia coli	Not Detected per 100 ml	Initial testing, then every year
Turbidity	Must not exceed 5 NTU (Nephelometric Turbidity Units)	Initial testing, then once every year



- Dairy: Test every supply of water before using it in food or drink production areas, and then test at least once each year. You must test as per Table 1, plus any additional water-use criteria that you have determined by completing Dairy Processors Template Water-use Plan.
- Dairy: Only use water tanks, containers, pipes, outlet taps and treatment systems for any water supplies on site that are suitable for drinking water (or are 'food-grade'). Regularly check and maintain these.
- Dairy: Clearly mark outlet taps, tanks, and pipes that do not contain clean water. These must not be used for food or drink processing, hand washing and cleaning.
- Dairy: You must have a system for managing cross contamination, dead ends and backflow.
- Dairy: **Records** are kept of all investigations and corrective actions.





Example scenarios	Actions
The town water supplier advises that the water is not fit for drinking without additional treatment (for example a boil water notice is issued)	The following actions are taken as appropriate to the scenario: 1. Immediate control and investigation of problem • all operations requiring the use of water are stopped; • the cause of the problem is investigated; • appropriate corrective actions are taken to rectify the problem (for example through further treatment); and • if necessary, an alternative source of suitable water is found. 2. What to do with affected product and equipment • any affected or potentially affected product is managed as per '<u>Taking action when something goes wrong</u>' [red] card; • any affected food contact surfaces are cleaned and sanitised prior to reuse; and • any affected packaging materials and containers that cannot be effectively cleaned and sanitised are not used for packaging of any product. 3. Further corrective actions • where product has been affected by the use of unsuitable water, reassessment of the water source is done, and actions put in place to mitigate the issue
Water does not meet any of the requirements and there are no other means to ensure the water is suitable at the point of use	
Water supply is contaminated by non-complying water	

Things to show your verifier

Your verifier will:

- ask how you know your water is fit for purpose,
- ask you about how you check and maintain water
- equipment and facilities (see the '[Maintaining equipment and facilities](#)' [med blue] card).



Dairy: Your verifier might ask to see:

- water reticulation plan (for example site plan)
- Dairy Processors Template Water-use Plan
- results of water testing
- results of ongoing monitoring of any water treatment activities (if applicable)
- documentation of reassessment
- any problems informed of or detected (for example notification from water supplier, failure of water treatment plant)
- any corrective action taken.

For self-supply water, your verifier will:

- ask to see test results for any roof, surface or ground water supplies that are used for food preparation, washing food contact surfaces/equipment or for hand washing ask what near-by activities could affect the safety of your water,
- ask if you have identified any hazards, and how you control these,
- ask you to show them how you know any water treatment system is working properly,
- ask about any chemical [hazards](#) you have identified and how you control these.



- Mobile businesses: Your verifier will ask how many water sources you use (for your different locations) and how you manage risks from water.